

1 Spin liquids

For reviews see [Savary and Balents 2017](#); [Zhou et al. 2017](#); [Lee et al. 2006](#); the presentation here follows [Wen \(2002, 2004\)](#).

A spin liquid is a ground state of an interacting spin system that breaks no symmetry, $\langle \mathbf{S}_i \rangle = 0$, down to zero temperature. Its characteristic properties are:

- (i) The elementary excitations carry fractional quantum numbers, e.g. neutral spin- $\frac{1}{2}$ excitations (spinons), whereas a local spin flip carries $S = 1$.
- (ii) Different spin liquids cannot, in general, be distinguished by their symmetry properties: two spin liquids with the same symmetry can be different phases. Spin liquids are examples of quantum disordered states, and the new type of order they possess is called *quantum order*.
- (iii) The ground states of gapped spin liquids have a topological degeneracy (e.g. on a torus) that cannot be related to any symmetry.
- (iv) Spin liquids always contain certain kinds of (emergent) gauge fluctuations.

Failure of the conventional mean-field theory. For the Heisenberg model

$$\mathcal{H} = \sum_{ij} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j, \quad (1.1)$$

the standard mean-field decoupling is

$$\mathcal{H}_{\text{mean}} = \sum_{ij} J_{ij} \left(\langle \mathbf{S}_i \rangle \cdot \mathbf{S}_j + \mathbf{S}_i \cdot \langle \mathbf{S}_j \rangle - \langle \mathbf{S}_i \rangle \cdot \langle \mathbf{S}_j \rangle \right), \quad (1.2)$$

with the self-consistency equation $\langle \mathbf{S}_i \rangle = \langle \Phi_{\text{mean}} | \mathbf{S}_i | \Phi_{\text{mean}} \rangle$, where $|\Phi_{\text{mean}}\rangle$ is the ground state of $\mathcal{H}_{\text{mean}}$. This approach can only describe ordered states, because one assumes from the beginning $\langle \mathbf{S}_i \rangle \neq 0$. To describe a spin liquid one needs a mean-field theory in which the order parameter is not $\langle \mathbf{S}_i \rangle$.

2 Fermionic parton construction

Write the spin- $\frac{1}{2}$ operator in terms of fermions (Abrikosov fermions ([Abrikosov, 1965](#)), or *spinons* ([Baskaran et al., 1987](#)))

$$\mathbf{S}_i = \frac{1}{2} f_{i\alpha}^\dagger \boldsymbol{\sigma}_{\alpha\beta} f_{i\beta}, \quad \alpha, \beta \in \{\uparrow, \downarrow\}. \quad (2.1)$$

The fermion Fock space at each site has four states $|0\rangle, |\uparrow\rangle, |\downarrow\rangle, |\uparrow\downarrow\rangle$, while the physical spin Hilbert space has only $|\uparrow\rangle, |\downarrow\rangle$. The physical states satisfy the constraints

$$f_{i\alpha}^\dagger f_{i\alpha} = 1, \quad f_{i\alpha} f_{i\beta} \epsilon_{\alpha\beta} = 0. \quad (2.2)$$

Heisenberg model in terms of spinons. Using $\boldsymbol{\sigma}_{\alpha\beta} \cdot \boldsymbol{\sigma}_{\gamma\delta} = 2\delta_{\alpha\delta}\delta_{\beta\gamma} - \delta_{\alpha\beta}\delta_{\gamma\delta}$ and the anticommutation relations ($i \neq j$),

$$\mathbf{S}_i \cdot \mathbf{S}_j = -\frac{1}{2} f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{i\beta} + \frac{1}{2} n_i - \frac{1}{4} n_i n_j, \quad n_i = f_{i\alpha}^\dagger f_{i\alpha}, \quad (2.3)$$

so that

$$\mathcal{H} = \sum_{ij} \left[-\frac{1}{2} J_{ij} f_{i\alpha}^\dagger f_{j\alpha} f_{j\beta}^\dagger f_{i\beta} + J_{ij} \left(\frac{1}{2} n_i - \frac{1}{4} n_i n_j \right) \right]. \quad (2.4)$$

The spin Hamiltonian and Eq. (2.4) are equivalent only when the constraint (2.2) is imposed.

Gauge redundancy. The physical spin operator is unchanged by the local phase transformation

$$f_{i\alpha} \rightarrow e^{i\theta_i} f_{i\alpha}. \quad (2.5)$$

This is a $U(1)$ gauge redundancy of the parton description. In fact the redundancy is larger (Affleck et al., 1988): in terms of the doublet $\psi_i = (f_{i\uparrow}, f_{i\downarrow}^\dagger)^T$ the spin operator is invariant under $\psi_i \rightarrow W_i \psi_i$ with $W_i \in SU(2)$, and the three constraints $\psi_i^\dagger \tau^l \psi_i = 0$ ($l = 1, 2, 3$) are exactly Eq. (2.2):

$$\psi_i^\dagger \tau^3 \psi_i = n_i - 1, \quad \psi_i^\dagger \tau^1 \psi_i = f_{i\uparrow}^\dagger f_{i\downarrow}^\dagger + f_{i\downarrow} f_{i\uparrow}. \quad (2.6)$$

3 Mean-field theory of spinons

The mean-field ground state (Baskaran et al., 1987; Affleck and Marston, 1988) is obtained at zeroth order by the following approximations.

1. The constraint is imposed only on average,

$$\langle f_{i\alpha}^\dagger f_{i\alpha} \rangle = 1. \quad (3.1)$$

It is enforced by including a site-dependent (and time-independent) Lagrange multiplier $a_0(i) (f_{i\alpha}^\dagger f_{i\alpha} - 1)$ in the Hamiltonian.

2. The operator $f_{i\alpha}^\dagger f_{j\alpha}$ is replaced by its ground-state expectation value

$$\chi_{ij} = \langle f_{i\alpha}^\dagger f_{j\alpha} \rangle, \quad \chi_{ji} = \chi_{ij}^*. \quad (3.2)$$

Then (dropping constants)

$$\mathcal{H}_{\text{mean}} = \sum_{\langle ij \rangle} -\frac{1}{2} J_{ij} \left[(f_{i\alpha}^\dagger f_{j\alpha} \chi_{ji} + \text{h.c.}) - |\chi_{ij}|^2 \right] + \sum_i a_0(i) (f_{i\alpha}^\dagger f_{i\alpha} - 1). \quad (3.3)$$

χ_{ij} is determined by the self-consistency equation (3.2), and the chemical potential $a_0(i)$ is chosen such that Eq. (3.1) is satisfied. The set $\{\chi_{ij}, a_0(i)\}$ is called the mean-field *ansatz*. Note that χ_{ij} and $a_0(i)$ do not change under spin rotations, so the mean-field state is a spin singlet.

Zeroth-order excitations. Let $\bar{\chi}_{ij}$, $\bar{a}_0(i)$ be a solution of Eqs. (3.1)–(3.2). If we ignore fluctuations of $\bar{\chi}_{ij}$ and $\bar{a}_0(i)$, the excitations around the mean-field state are described by

$$\mathcal{H}_{\text{mean}} = \sum_{\langle ij \rangle} -\frac{J_{ij}}{2} \left[(f_{i\alpha}^\dagger f_{j\alpha} \bar{\chi}_{ji} + \text{h.c.}) - |\bar{\chi}_{ij}|^2 \right] + \sum_i \bar{a}_0(i) (f_{i\alpha}^\dagger f_{i\alpha} - 1). \quad (3.4)$$

These are free spinons, created by $f_{i\alpha}^\dagger$: neutral spin- $\frac{1}{2}$ fermions.

The question is whether we should trust this result: should we believe in the existence of spin liquids with neutral spin- $\frac{1}{2}$ fermionic excitations? One way to check this is to include the fluctuations and see whether they change the mean-field result.

4 Beyond mean field: gauge fluctuations

The fluctuations of χ_{ij} and $a_0(i)$ describe the collective excitations above the mean-field ground state. χ_{ij} has two kinds of fluctuations, amplitude and phase,

$$\chi_{ij} = \bar{\chi}_{ij} e^{-ia_{ij}}. \quad (4.1)$$

The amplitude fluctuations have a finite energy gap and can be ignored at low energy. Keeping the phase fluctuations a_{ij} and $a_0(i)$ gives the first-order mean-field Hamiltonian

$$\mathcal{H} = \sum_{\langle ij \rangle} -\frac{J_{ij}}{2} \left(f_{i\alpha}^\dagger f_{j\alpha} \bar{\chi}_{ji} e^{-ia_{ji}} + \text{h.c.} \right) + \sum_i a_0(i) (f_{i\alpha}^\dagger f_{i\alpha} - 1). \quad (4.2)$$

Eq. (4.2) is invariant under the local transformation

$$f_i \rightarrow e^{i\theta_i} f_i, \quad a_{ij} \rightarrow a_{ij} + \theta_i - \theta_j, \quad (4.3)$$

($a_0(i) \rightarrow a_0(i) + \partial_t \theta_i$ for time-dependent θ_i). Thus a_{ij} and $a_0(i)$ are the spatial and time components of a $U(1)$ lattice gauge field, and in first-order mean-field theory the excitations are spinons coupled to a $U(1)$ gauge field (instead of free spinons in the zeroth-order theory). The gauge fluctuations can drastically change the properties of the spinons.

How can a spin model have gauge fields and fermions? The original model is an interacting spin model, which is a purely bosonic model. We first split the bosonic spin operator into two fermionic operators, and then introduce the gauge field to glue them back into bosonic spins. One may then expect that the gauge bosons and fermions are artefacts of the construction, and that at the end all we get are bosonic spin fluctuations. Whether this is so depends on the dynamics of the gauge field:

- **Confining phase.** The spinons interact through a linear potential and never appear as quasiparticles. The gauge bosons have a large energy gap and do not appear in the low-energy spectrum. The only low-energy excitations are spinon pairs, i.e. bosonic spin fluctuations. (In $2+1$ dimensions a compact $U(1)$ gauge field without gapless matter is confining (Polyakov, 1977).)
- **Deconfined phase.** The spinons and the gauge bosons appear as quasiparticles. The low-energy theory is then really a theory of fermions and gauge fields: the spin liquid is a new phase.

Example: the dimer state. The first-order mean-field theory of the dimer state has a confining $U(1)$ gauge interaction. Only spinon bound states appear as physical excitations.

Mean-field theory with a changed Hilbert space. In the usual mean-field theory we approximate the interacting Hamiltonian in order to simplify the problem; the Hilbert space is not changed. In the parton construction the Hilbert space *is* changed: the mean-field state $|\Psi_{\text{mean}}\rangle$ is not even a valid spin wavefunction, since it contains sites with zero or two fermions. Including the gauge fluctuations around the mean-field state is what restores the constraint.

5 Projective construction

A physical spin wavefunction has exactly one fermion per site. A valid spin state is obtained from the mean-field state by Gutzwiller projection,

$$\Psi_{\text{spin}}^{\chi_{ij}}(\{\alpha_i\}) = \langle 0_f | \prod_i f_{i\alpha_i} | \Psi_{\text{mean}}^{\chi_{ij}} \rangle, \quad (5.1)$$

where α_i is the spin at site i and $|0_f\rangle$ the fermion vacuum.

Eq. (5.1) gives the gauge transformation a physical meaning: two ansatz related by

$$\tilde{\chi}_{ij} = e^{i\theta_i} \chi_{ij} e^{-i\theta_j} \quad (5.2)$$

give rise to the same projected spin state,

$$\Psi_{\text{spin}}^{\tilde{\chi}_{ij}}(\{\alpha_i\}) = e^{i\sum_i \theta_i} \Psi_{\text{spin}}^{\chi_{ij}}(\{\alpha_i\}), \quad (5.3)$$

which differs only by a global phase. On the other hand, choices of χ_{ij} that are not gauge equivalent give different mean-field wavefunctions $|\Psi_{\text{mean}}^{\chi_{ij}}\rangle$ and, after projection, different spin states. A spin liquid is therefore labelled by a gauge equivalence class of ansatz.

6 Schwinger-boson mean-field theory

Alternatively, represent the spin by Schwinger bosons ([Arovas and Auerbach, 1988](#); [Auerbach, 1994](#); [Sachdev, 1992](#)),

$$\mathbf{S}_i = \frac{1}{2} b_{i\sigma}^\dagger \boldsymbol{\sigma}_{\sigma\sigma'} b_{i\sigma'}, \quad \mathbf{S}_i \cdot \mathbf{S}_j = \frac{1}{4} (b_{i\sigma}^\dagger \boldsymbol{\sigma}_{\sigma\sigma'} b_{i\sigma'}) \cdot (b_{j\tau}^\dagger \boldsymbol{\sigma}_{\tau\tau'} b_{j\tau'}), \quad (6.1)$$

with the constraint that at every site

$$\sum_{\sigma} b_{i\sigma}^\dagger b_{i\sigma} = \kappa \quad (\kappa = 2S). \quad (6.2)$$

Define the bond operators

$$\hat{A}_{ij} = \frac{1}{2} \sum_{\sigma\sigma'} \epsilon_{\sigma\sigma'} b_{i\sigma} b_{j\sigma'}, \quad \hat{B}_{ij} = \frac{1}{2} \sum_{\sigma} b_{i\sigma}^\dagger b_{j\sigma}. \quad (6.3)$$

$\hat{A}_{ij}$ creates/annihilates a singlet on the bond (antiferromagnetic correlations), $\hat{B}_{ij}$ describes hopping (ferromagnetic correlations). Within the physical Hilbert space,

$$\mathbf{S}_i \cdot \mathbf{S}_j = :\hat{B}_{ij}^\dagger \hat{B}_{ij}: - \hat{A}_{ij}^\dagger \hat{A}_{ij}. \quad (6.4)$$

Decoupling in $\hat{A}$ and $\hat{B}$:

$$\mathcal{H}_{\text{MF}} = \sum_{\langle ij \rangle} J_{ij} \left[(B_{ij}^* \hat{B}_{ij} + \text{h.c.}) - (A_{ij}^* \hat{A}_{ij} + \text{h.c.}) - |B_{ij}|^2 + |A_{ij}|^2 \right] + \mu \sum_i \left(\sum_{\sigma} b_{i\sigma}^\dagger b_{i\sigma} - \kappa \right). \quad (6.5)$$

The mean-field Hamiltonian is used to generate $|\Psi(A_{ij}, B_{ij}, \mu)\rangle$, which should then be projected onto the constrained Hilbert space. The hard projection is difficult to implement analytically, so we impose the constraint not locally but on average,

$$\sum_{\sigma} \langle b_{i\sigma}^\dagger b_{i\sigma} \rangle = \kappa, \quad (6.6)$$

and solve the self-consistency equations

$$\langle \hat{A}_{ij} \rangle = A_{ij}, \quad \langle \hat{B}_{ij} \rangle = B_{ij}. \quad (6.7)$$

Schwinger-boson mean-field states are either magnetically ordered (when the spinon gap closes and the bosons condense) or gapped spin liquids; gapless spin liquids are naturally described by fermionic partons.

Gauge structure. $\mathcal{H}_{\text{MF}}$ is invariant under

$$b_{i\sigma} \rightarrow e^{i\theta_i} b_{i\sigma}, \quad A_{ij} \rightarrow e^{i(\theta_i + \theta_j)} A_{ij}, \quad B_{ij} \rightarrow e^{-i(\theta_i - \theta_j)} B_{ij}. \quad (6.8)$$

For bosons the local gauge transformation is $U(1)$ (not $SU(2)$, as in the case of fermions). A nonzero pairing A_{ij} on bonds that form odd loops (e.g. the triangular lattice) leaves only $\theta_i \in \{0, \pi\}$ uniform: the remaining gauge group is $\mathbb{Z}_2$, i.e. a $\mathbb{Z}_2$ spin liquid.

7 The projective symmetry group

Motivation

For the mean-field theory to survive projection (equivalently, to survive fluctuations) and give rise to a spin-liquid state, we require that it reflects the microscopic symmetries of the spin model. This leads to *symmetric spin liquids* (Wen, 2002). The symmetry transformations include the space group, spin rotation and time reversal.

A symmetry transformation need not leave the ansatz invariant: it suffices that it maps the ansatz to a gauge-transformed form, since the same spin wavefunction is obtained after projection. Hence a symmetry transformation that leaves the ansatz invariant is in general a naive transformation combined with a local gauge transformation.

The **projective symmetry group** (PSG) of an ansatz is the set of all transformations that leave the ansatz invariant,

$$\text{PSG} = \{ G_U U \mid G_U U(u_{ij}) = u_{ij} \}, \quad (7.1)$$

where U is a symmetry transformation and G_U a gauge transformation. The PSG characterizes the quantum order of a spin liquid; it is a property of an ansatz.

For fermionic partons in the $SU(2)$ formulation the ansatz is the set of 2×2 matrices u_{ij} in $\mathcal{H}_{\text{mean}} = -\sum_{\langle ij \rangle} (\psi_i^\dagger u_{ij} \psi_j + \text{h.c.}) + \dots$, and

$$U(u_{ij}) = u_{U(i), U(j)} \quad (\text{e.g. translation}), \quad (7.2)$$

$$G_U(u_{ij}) = G_U(i) u_{ij} G_U^\dagger(j), \quad G_U(i) \in SU(2). \quad (7.3)$$

For Schwinger bosons the gauge transformations are phases, $G_U(i) = e^{i\phi_U(i)}$, acting on (A_{ij}, B_{ij}) as in Sec. 6.

Invariant gauge group

The PSG contains as a special subgroup the pure gauge transformations that leave the ansatz invariant, the *invariant gauge group* (IGG),

$$\mathcal{G} = \{ W_i \mid W_i u_{ij} W_j^\dagger = u_{ij}, \ W_i \in SU(2) \}. \quad (7.4)$$

These elements are not the result of any physical symmetry. It is therefore natural to associate the IGG with the emergent gauge group of the low-energy theory of the spin-liquid phase ($\mathbb{Z}_2$, $U(1)$ or $SU(2)$ spin liquids). If $\mathcal{G}$ is nontrivial, then for a fixed symmetry transformation U there are many gauge transformations G_U that leave the ansatz invariant, and

$$\text{SG} = \text{PSG}/\mathcal{G}. \quad (7.5)$$

The PSG is a projective representation (an extension) of the symmetry group.

Gauge transformation of a PSG

Consider two gauge-equivalent ansatz, u_{ij} and $W(u_{ij}) = W(i)u_{ij}W^\dagger(j)$. If $G_U U \in \text{PSG}(u_{ij})$, then

$$(WG_U U W^{-1}) W(u_{ij}) = W G_U U(u_{ij}) = W(u_{ij}), \quad (7.6)$$

and $WG_U U W^{-1} = (WG_U W_U^{-1})U$ with $W_U \equiv UWU^{-1}$ (the gauge transformation W with its argument moved by U). Hence

$$\boxed{G_U \rightarrow W G_U W_U^{-1}} \quad (7.7)$$

is in the PSG of the gauge-transformed ansatz. PSGs related in this way describe the same phase; the classification is of PSGs modulo this transformation.

Algebraic constraints

Let the symmetry group be generated by $X_1, X_2, \dots$ with relations $X_{a_1}^{\pm 1} X_{a_2}^{\pm 1} \dots = \mathbb{I}$. The same product of PSG elements need not be the identity, but it must be a pure gauge transformation that leaves the ansatz invariant:

$$(G_{X_{a_1}} X_{a_1})^{\pm 1} (G_{X_{a_2}} X_{a_2})^{\pm 1} \dots \in \mathcal{G}. \quad (7.8)$$

These *algebraic constraints*, solved modulo gauge transformations, give the classification of PSGs. Useful identities: for a symmetry X and a gauge transformation G ,

$$XGX^{-1} = G(X^{-1}(i)), \quad (7.9)$$

i.e. conjugation by X moves the argument of the gauge transformation.

8 Example: $\mathbb{Z}_2$ PSGs of translations (fermionic partons)

Let the IGG be

$$\mathcal{G} = \mathbb{Z}_2 = \{G_1, G_2\}, \quad G_1(i) = \tau^0, \quad G_2(i) = -\tau^0, \quad (8.1)$$

and let the symmetry group be the translations T_x, T_y of the square lattice,

$$T_x(u_{ij}) = u_{i-\hat{x}, j-\hat{x}}, \quad T_y(u_{ij}) = u_{i-\hat{y}, j-\hat{y}}. \quad (8.2)$$

The PSG is generated by $\{\pm G_x T_x, \pm G_y T_y\}$.

Step 1: uniform gauge. Due to the translation symmetry of the ansatz we can choose a gauge in which all the loop operators $P_C = u_{ij}u_{jk} \cdots u_{li}$ are translation invariant: $P_{C_1} = P_{C_2}$ if the loops C_1, C_2 are related by a translation. Under $G_x T_x$ a loop operator based at i transforms as

$$P_C \rightarrow G_x(i') P_{T_x C} G_x^\dagger(i') = G_x(i') P_C G_x^\dagger(i'), \quad i' = T_x i. \quad (8.3)$$

Translation invariance of P_C therefore requires that $G_x(i)$ commutes with all loop operators based at i . For a $\mathbb{Z}_2$ spin liquid these loop operators do not all commute with each other, so

$$G_x(i) = \pm \tau^0, \quad G_y(i) = \pm \tau^0. \quad (8.4)$$

A further gauge transformation $W(i) = \pm \tau^0$ does not change the loop operators. We use it to set $G_x(i) = \tau^0$.

Step 2: algebraic constraint. The translations commute, $T_x T_y T_x^{-1} T_y^{-1} = \mathbb{I}$, so for any ansatz

$$G_x T_x G_y T_y (G_x T_x)^{-1} (G_y T_y)^{-1} = G_x(i) G_y(i - \hat{x}) G_x^{-1}(i - \hat{y}) G_y^{-1}(i) \in \mathcal{G}, \quad (8.5)$$

where we used $T_x G_y T_x^{-1} = G_y(i - \hat{x})$ and $T_y G_x^{-1} T_y^{-1} = G_x^{-1}(i - \hat{y})$. With $G_x = \tau^0$:

$$G_y(i - \hat{x}) G_y^{-1}(i) = \eta \tau^0, \quad \eta = \pm 1. \quad (8.6)$$

Hence $G_y(i) = \eta^{i_x} G_y(0, i_y)$, and a gauge transformation depending only on i_y (which keeps $G_x = \tau^0$) sets $G_y(0, i_y) = \tau^0$.

There are two $\mathbb{Z}_2$ PSGs for translations:

$$G_x(i) = \tau^0, \quad G_y(i) = \eta^{i_x} \tau^0, \quad \begin{cases} \eta = +1 : & \mathbb{Z}2A, \\ \eta = -1 : & \mathbb{Z}2B. \end{cases} \quad (8.7)$$

For $\mathbb{Z}2B$ the translations anticommute when acting on spinons, $T_x T_y T_x^{-1} T_y^{-1} = -1$: the spinons see a π flux per unit cell.

Remark. The names $\mathbb{Z}2A$ and $\mathbb{Z}2B$ are those of the classification of [Wen \(2002\)](#) of $\mathbb{Z}_2$ spin liquids on the square lattice. Wen writes the $\mathbb{Z}2B$ PSG as $G_x(i) = (-)^{i_y} \tau^0$, $G_y(i) = \tau^0$. This is the same PSG in a different gauge: the transformation $G_U \rightarrow W G_U W_U^{-1}$ with $W(i) = (-)^{i_x i_y} \tau^0$ maps $\{G_x = \tau^0, G_y = (-)^{i_x} \tau^0\}$ to $\{G_x = (-)^{i_y} \tau^0, G_y = \tau^0\}$. The gauge-invariant content is the sign η of $T_x T_y T_x^{-1} T_y^{-1}$ acting on the spinons.

9 Example: $\mathbb{Z}_2$ PSGs of Schwinger bosons on the triangular lattice

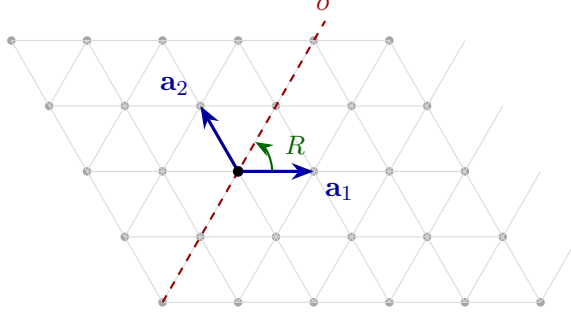
This classification was carried out by [Wang and Vishwanath \(2006\)](#); see also [Messio et al. \(2013\)](#).

Lattice and symmetry generators

Write $\mathbf{r} = r_1 \mathbf{a}_1 + r_2 \mathbf{a}_2$ with $\mathbf{a}_1 = (1, 0)$, $\mathbf{a}_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$. The space group is generated by

$$\begin{aligned} T_1 : (r_1, r_2) &\rightarrow (r_1 + 1, r_2), & T_2 : (r_1, r_2) &\rightarrow (r_1, r_2 + 1), \\ \sigma : (r_1, r_2) &\rightarrow (r_2, r_1), & R_{\pi/3} : (r_1, r_2) &\rightarrow (r_1 - r_2, r_1), \end{aligned} \quad (9.1)$$

where σ is the reflection about the line along $\mathbf{a}_1 + \mathbf{a}_2$ and $R_{\pi/3}$ the rotation by 60° about a site. We write $R \equiv R_{\pi/3}$; $R^{-1} : (r_1, r_2) \rightarrow (r_2, r_2 - r_1)$.



Composition rule

Let $G_X : b_i \rightarrow e^{i\phi_X(i)} b_i$. Acting on a boson operator,

$$G_X X : b_{\mathbf{r}} \rightarrow e^{i\phi_X(X(\mathbf{r}))} b_{X(\mathbf{r})}, \quad (G_X X)^{-1} : b_{\mathbf{r}} \rightarrow e^{-i\phi_X(\mathbf{r})} b_{X^{-1}(\mathbf{r})}. \quad (9.2)$$

For a product of PSG elements acting on $b_{\mathbf{r}}$ (rightmost first), each factor $G_X X$ contributes $+\phi_X$ evaluated at the site *after* it acts, and each factor $(G_X X)^{-1}$ contributes $-\phi_X$ evaluated at the site *before* it acts. With $\mathcal{G} = \mathbb{Z}_2 = \{1, -1\}$ every algebraic constraint (7.8) becomes an equation for the phases,

$$\sum (\text{phases}) = p\pi, \quad p \in \{0, 1\} \pmod{2\pi}. \quad (9.3)$$

Note also that ϕ_X is defined only up to a constant π , since G_X and $-G_X$ differ by an element of the IGG.

Under a gauge transformation $G : b_i \rightarrow e^{i\phi_G(i)} b_i$ the PSG transforms as $G_X X \rightarrow G G_X X G^{-1}$, i.e.

$$\phi_X(\mathbf{r}) \rightarrow \phi_G(\mathbf{r}) + \phi_X(\mathbf{r}) - \phi_G(X^{-1}(\mathbf{r})). \quad (9.4)$$

Translations

The relation $T_1^{-1} T_2 T_1 T_2^{-1} = \mathbb{I}$ gives

$$(G_{T_1} T_1)^{-1} (G_{T_2} T_2) (G_{T_1} T_1) (G_{T_2} T_2)^{-1} \in \mathcal{G}, \quad (9.5)$$

or, in terms of phases,

$$-\phi_{T_1}(T_1(\mathbf{r})) + \phi_{T_2}(T_1(\mathbf{r})) + \phi_{T_1}(T_1 T_2^{-1}(\mathbf{r})) - \phi_{T_2}(\mathbf{r}) = p_1 \pi. \quad (9.6)$$

Gauge fixing. Using Eq. (9.4) with $X = T_1$, we choose ϕ_G such that $\phi_G(r_1, r_2) - \phi_G(r_1 - 1, r_2) = -\phi_{T_1}(r_1, r_2)$. This makes

$$\phi_{T_1}(\mathbf{r}) = 0. \quad (9.7)$$

Eq. (9.6) then reads $\phi_{T_2}(r_1 + 1, r_2) - \phi_{T_2}(r_1, r_2) = p_1 \pi$, so

$$\phi_{T_2}(r_1, r_2) = \phi_{T_2}(0, r_2) + p_1 \pi r_1. \quad (9.8)$$

A further gauge transformation depending only on r_2 does not change $\phi_{T_1} = 0$, and sets $\phi_{T_2}(0, r_2) = 0$. Any further gauge choice must not change the previous choices. We obtain

$$\phi_{T_1}(r_1, r_2) = 0, \quad \phi_{T_2}(r_1, r_2) = p_1 \pi r_1. \quad (9.9)$$

(We consider an infinite lattice, or open boundary conditions; on a torus $p_1 = 1$ requires an even number of sites along $\mathbf{a}_1$.)

Remaining gauge freedom. After this gauge fixing we are still left with three gauge transformations that keep this form of ϕ_{T_1}, ϕ_{T_2} (up to the constant π allowed by the IGG):

$$G_1 : \phi_1(\mathbf{r}) = \text{const}, \quad G_2 : \phi_2(\mathbf{r}) = \pi r_1, \quad G_3 : \phi_3(\mathbf{r}) = \pi r_2. \quad (9.10)$$

By Eq. (9.4), G_1 does not change any generator. G_2 changes $\phi_{T_1} \rightarrow \phi_{T_1} + \pi$ and leaves ϕ_{T_2} unchanged; G_3 leaves ϕ_{T_1} unchanged and changes $\phi_{T_2} \rightarrow \phi_{T_2} + \pi$. These constant shifts are allowed because of the IGG. G_2 and G_3 will, however, modify the other generators of the PSG.

Reflection σ

The algebraic relations are

$$T_1^{-1} \sigma T_2 \sigma^{-1} = \mathbb{I}, \quad T_2^{-1} \sigma T_1 \sigma^{-1} = \mathbb{I}, \quad \sigma^2 = \mathbb{I}. \quad (9.11)$$

In terms of phases the first two give

$$\phi_\sigma(r_1 + 1, r_2) - \phi_\sigma(r_1, r_2) = p'_2 \pi + p_1 \pi r_2, \quad (9.12)$$

$$\phi_\sigma(r_1, r_2 + 1) - \phi_\sigma(r_1, r_2) = p'_3 \pi + p_1 \pi r_1, \quad (9.13)$$

(signs of multiples of π are irrelevant mod 2π), with solution

$$\phi_\sigma(r_1, r_2) = \phi_\sigma(0, 0) + p'_2 \pi r_1 + p'_3 \pi r_2 + p_1 \pi r_1 r_2. \quad (9.14)$$

A further constraint comes from $\sigma\sigma = \mathbb{I}$: $G_\sigma \sigma G_\sigma \sigma \in \mathcal{G}$, i.e.

$$\phi_\sigma(r_1, r_2) + \phi_\sigma(r_2, r_1) = p_2 \pi. \quad (9.15)$$

At $r_1 = r_2 = 0$: $\phi_\sigma(0, 0) = p_2 \pi / 2$. For general $\mathbf{r}$: $(p'_2 + p'_3) \pi (r_1 + r_2) = 0 \text{ mod } 2\pi$, so

$$p'_2 = p'_3 \pmod{2}. \quad (9.16)$$

Under the gauge transformation G_2 ,

$$\phi_\sigma(r_1, r_2) \rightarrow \phi_\sigma(r_1, r_2) + \pi r_1 - \pi r_2 \equiv \phi_\sigma(r_1, r_2) + \pi(r_1 + r_2), \quad (9.17)$$

which shifts p'_2 and p'_3 by one. Hence we can set $p'_2 = p'_3 = 0$:

$$\phi_\sigma(r_1, r_2) = \frac{p_2 \pi}{2} + p_1 \pi r_1 r_2. \quad (9.18)$$

The combination $G_2 G_3$, $\phi_{23}(\mathbf{r}) = \pi(r_1 + r_2)$, leaves ϕ_{T_1}, ϕ_{T_2} unchanged up to the constant π and also leaves ϕ_σ unchanged: $\phi_\sigma \rightarrow \phi_\sigma + \pi(r_1 + r_2) - \pi(r_2 + r_1) = \phi_\sigma$.

Rotation $R_{\pi/3}$

The algebraic relations with the translations are

$$T_1^{-1} R T_2^{-1} R^{-1} = \mathbb{I}, \quad T_2^{-1} R T_2 T_1 R^{-1} = \mathbb{I}, \quad (9.19)$$

as follows from $R T_2^{-1} R^{-1} = T_1$ and $R T_2 T_1 R^{-1} = T_2$.

First relation. $(G_{T_1}T_1)^{-1}(G_R R)(G_{T_2}T_2)^{-1}(G_R R)^{-1} \in \mathcal{G}$ gives

$$-\phi_{T_1}(T_1(\mathbf{r})) + \phi_R(T_1(\mathbf{r})) - \phi_{T_2}(R^{-1}(\mathbf{r})) - \phi_R(\mathbf{r}) = p'_4\pi. \quad (9.20)$$

With $\phi_{T_1} = 0$ and $\phi_{T_2}(R^{-1}\mathbf{r}) = \phi_{T_2}(r_2, r_2 - r_1) = p_1\pi r_2$,

$$\phi_R(r_1 + 1, r_2) - \phi_R(r_1, r_2) = p'_4\pi + p_1\pi r_2, \quad (9.21)$$

so

$$\phi_R(r_1, r_2) = \phi_R(0, r_2) + p'_4\pi r_1 + p_1\pi r_1 r_2. \quad (9.22)$$

Second relation. $(G_{T_2}T_2)^{-1}(G_R R)(G_{T_2}T_2)(G_{T_1}T_1)(G_R R)^{-1} \in \mathcal{G}$ gives

$$-\phi_{T_2}(T_2(\mathbf{r})) + \phi_R(T_2(\mathbf{r})) + \phi_{T_2}(T_2T_1R^{-1}(\mathbf{r})) - \phi_R(\mathbf{r}) = p'_5\pi, \quad (9.23)$$

with $T_2T_1R^{-1}(\mathbf{r}) = (r_2 + 1, r_2 - r_1 + 1)$. Therefore

$$\phi_R(r_1, r_2 + 1) - \phi_R(r_1, r_2) = p'_5\pi + p_1\pi(r_1 - r_2 - 1). \quad (9.24)$$

At $r_1 = 0$ this is a recursion in r_2 ; summing it,

$$\phi_R(0, r_2) = \phi_R(0, 0) + p'_5\pi r_2 - p_1\pi \sum_{k=1}^{r_2} k = \phi_R(0, 0) + p'_5\pi r_2 - p_1\pi \frac{r_2(r_2 + 1)}{2}. \quad (9.25)$$

Since $-\frac{1}{2}r_2(r_2 + 1) \equiv \frac{1}{2}r_2(r_2 - 1) + r_2 \pmod{2}$, absorbing $p_1\pi r_2$ into $p'_5\pi r_2$ gives

$$\phi_R(r_1, r_2) = \phi_R(0, 0) + p'_4\pi r_1 + p'_5\pi r_2 + p_1\pi \left[r_1 r_2 + \frac{r_2(r_2 - 1)}{2} \right]. \quad (9.26)$$

where now Eq. (9.24) holds with $p'_5 \rightarrow p'_5 + p_1$.

Point-group relations. The remaining relations are

$$(R\sigma)^2 = \mathbb{I}, \quad R^6 = \mathbb{I}. \quad (9.27)$$

For $(R\sigma)^2$, with $\sigma(\mathbf{r}) = (r_2, r_1)$, $R\sigma(\mathbf{r}) = (r_2 - r_1, r_2)$ and $\sigma R\sigma(\mathbf{r}) = (r_2, r_2 - r_1)$:

$$\begin{aligned} & \phi_\sigma(\sigma(\mathbf{r})) + \phi_R(R\sigma(\mathbf{r})) + \phi_\sigma(\sigma R\sigma(\mathbf{r})) + \phi_R(\mathbf{r}) \\ &= 2\phi_R(0, 0) + p_2\pi + p'_4\pi r_2 + p_1\pi[3r_2^2 - r_2] = 2\phi_R(0, 0) + p_2\pi + p'_4\pi r_2 \pmod{2\pi}. \end{aligned} \quad (9.28)$$

This must be a constant multiple of π , so $p'_4 = 0$ and $2\phi_R(0, 0) \in \{0, \pi\} \pmod{2\pi}$, i.e. $\phi_R(0, 0) = p_3\pi/2$ (using the IGG freedom $\phi_R \rightarrow \phi_R + \pi$). For R^6 , the orbit of $\mathbf{r}$ under R is

$$(r_1, r_2) \rightarrow (r_1 - r_2, r_1) \rightarrow (-r_2, r_1 - r_2) \rightarrow (-r_1, -r_2) \rightarrow (r_2 - r_1, -r_1) \rightarrow (r_2, r_2 - r_1) \rightarrow (r_1, r_2), \quad (9.29)$$

and summing ϕ_R over the orbit, the p'_5 and p_1 terms cancel mod 2π , leaving $6\phi_R(0, 0) = p_3\pi \pmod{2\pi}$: no new constraint.

Finally, the gauge transformation G_2G_3 changes

$$\phi_R(\mathbf{r}) \rightarrow \phi_R(\mathbf{r}) + \pi(r_1 + r_2) - \pi(r_2 + (r_2 - r_1)) \equiv \phi_R(\mathbf{r}) + \pi r_2, \quad (9.30)$$

while leaving $\phi_{T_1}, \phi_{T_2}, \phi_\sigma$ unchanged (up to the constants allowed by the IGG). Hence we can set $p'_5 = 0$.

Result

The $\mathbb{Z}_2$ PSGs of Schwinger-boson mean-field states on the triangular lattice are

$$\begin{aligned}\phi_{T_1}(r_1, r_2) &= 0, \\ \phi_{T_2}(r_1, r_2) &= p_1 \pi r_1, \\ \phi_{\sigma}(r_1, r_2) &= \frac{p_2 \pi}{2} + p_1 \pi r_1 r_2, \\ \phi_{R_{\pi/3}}(r_1, r_2) &= \frac{p_3 \pi}{2} + p_1 \pi \left[r_1 r_2 + \frac{r_2(r_2 - 1)}{2} \right],\end{aligned}\tag{9.31}$$

with $p_1, p_2, p_3 \in \{0, 1\}$: $2^3 = 8$ algebraic PSGs.

Physical meaning of p_1 . p_1 is the IGG element in $T_1^{-1}T_2T_1T_2^{-1}$: for $p_1 = 1$ the translations anticommute when acting on the spinons, which therefore see a π flux per unit cell (the analogue of $\mathbb{Z}2B$ above). $p_1 = 0$ gives the zero-flux states.

Which PSGs allow a nearest-neighbour ansatz? An ansatz is invariant under $G_X X$ if

$$A_{X(i)X(j)} = e^{i[\phi_X(X(i)) + \phi_X(X(j))]} A_{ij}, \quad B_{X(i)X(j)} = e^{i[\phi_X(X(j)) - \phi_X(X(i))]} B_{ij},\tag{9.32}$$

together with $A_{ji} = -A_{ij}$ and $B_{ji} = B_{ij}^*$. Starting from one nearest-neighbour bond and generating all other bonds with the PSG, the nearest-neighbour pairing A_{ij} is consistent (nonzero) for only two of the eight PSGs:

p_1	p_2	p_3	state
0	0	1	zero-flux $\mathbb{Z}_2$ spin liquid
1	1	0	π -flux $\mathbb{Z}_2$ spin liquid

For all other PSGs the symmetry forces $A_{ij} = 0$ on nearest-neighbour bonds, and a $\mathbb{Z}_2$ spin liquid with that PSG needs longer-range pairing. Thus, among nearest-neighbour Schwinger-boson states, the triangular lattice has exactly two symmetric $\mathbb{Z}_2$ spin liquids.

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